

Applied Topology in Albany (ATiA) Seminar

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“PERCEPTRONS” REVISITED: LEARNING HOMOLOGY AND THE EULER CHARACTERISTIC

Friday, October 2nd, 2026
12:00 p.m. Hudson 0110

ABSTRACT. Minsky and Papert’s landmark 1969 text “Perceptrons” is often blamed for causing the first AI winter because it showed that a single layer neural net can only learn the Euler characteristic of a subset of the plane. In this talk we will review some of the impossibility theorems for learning topological features and then transition to testing these ideas computationally. This will involve a brief detour in random topology for binary image data and then a comparison of what a 2-layer MLP can learn. We’ll show results for predicting—without using homology— β_0 , β_1 , and χ for all 5×5 grid pixel images. If time permits, we will discuss whether neural nets can learn Euler’s formula from the latent representations of these images alone.